Exploratory analysis (unsupervised data mining technique) using various clustering methods:

In this report, we conducted an **Exploratory analysis** (unsupervised techniques) on the provided dataset, which focuses on assessing the risk of heart disease based on key health indicators. Our primary approach involved applying unsupervised clustering methods. As mentioned, that during the data preparation process, we encountered the class imbalance for which we use sample node and further focused on stratification to improve this situation. This data balancing process was crucial to ensure that our subsequent analysis would be accurate and effective. For our analysis, we employed three distinct clustering methods: Ward Clustering, Average Clustering, and Centroid Clustering. In parallel, we also worked with a 90% sample of the original dataset to assess data stability. This dual approach allowed us to compare the clusters generated by each method with their respective 90% sample cluster nodes, providing insights into data consistency and reliability.

Ward's method, average linkage, and centroid linkage are hierarchical clustering methods that build a hierarchy of clusters by merging clusters until there is only one cluster left (James et al., 2013; Kaufman & Rousseeuw, 1990). This process helps uncover hidden insights, identify relationships, and segment the data into meaningful groups (Kaufman & Rousseeuw, 1990). Clustering stands as a valuable method applicable across diverse domains, including data mining, market segmentation, customer characterization, and image analysis, to name a few.

The further analysis is on Ward Clustering on the basis of most stable clusters. There are 3 distinct clusters. Cluster Segment ID 2 exhibits the highest values among these clusters. In the mean statistics window, Cluster 2 stands out with the most frequency records, around 96,232, while Cluster 1 has the lowest frequency with just 9,058 records, making it the least populated cluster in Ward clustering. When we compare variables across the clusters, Cluster 1 claims the highest BMI rate among all clusters, with a BMI value of 29.72298 and Cluster 2 shows the lowest BMI rate, recording 27.51647. But Cluster 2 takes the top spot for the "Age_Category = Adults" variable, accounting for 72.2%, and it ranks lowest for "Age_Category = Older Adults" at 27.8%. Cluster 3, on the other hand, displays the lowest rate for "Age_Category = Adults" at approximately 26.8% and the highest rate for "Age_Category = Older Adults" at 73.2%.

Moreover, Cluster 2 indicates a high percentage of "non-diabetic" individuals, at 90.6%, with only 7.5% falling into the "Diabetic = yes" category, while a little rate of only 1.9% are categorized as "Diabetic = borderline."

Turning our attention to the variable "smoking". both Cluster 1 and Cluster 3 reveal higher percentages of smokers, with rates of 54.9% and 56%, respectively, whereas 45.1% and 44% of individuals in these clusters do not smoke, respectively. An interesting observation emerges regarding "physical activity," as all clusters (1, 2, and 3) exhibit higher rates of "Physical Activity = Yes," with percentages of 54.1%, 84.7%, and 65.2%, respectively. This suggests that the dataset contains a substantial number of physically active individuals.

For a more complete understanding of the factors contributing to heart disease, we can turn to variable importance. It becomes clear that BMI is the most crucial variable, assigned a value of 1, signifying its paramount significance in assessing the risk of heart disease. Higher BMI values are associated with an increased likelihood of heart disease. In contrast, smoking is deemed the least important variable, with a value of 0.024322. Where gender holds the last position with a value of 0, indicating its insignificance in the context of heart disease risk assessment. These findings shed light on the factors and variable values supporting the risk of heart disease.

90% Ward Clustering analysis, which we conducted on a 90% sample of the original dataset, resulted in four distinct clusters. Cluster Segment ID 4 stands out with the highest frequency rate. Let's dig into the mean statistics to gain a better understanding of the insights resulting from this 90% sampling.

As observed, Cluster 4 contains the most frequency records, around 73,012, while Cluster 3 holds the least, with only 6,872 records. When we examine the BMI values, Cluster 1 leads with a BMI rate of 30.18256, suggesting a higher risk of heart disease. Cluster 4, however, lags slightly behind with a BMI rate of 27.59631, in close proximity to Cluster 3, which records a BMI rate of 27.75886.

Regarding the "Age_Category" variable, Clusters 1 and 2 have a higher proportion of older adults, comprising 66.9% and 85.4% of their populations, respectively. In contrast, Clusters 3 and 4 have a higher percentage of adults, with proportions of 58.4% and 84.6%, respectively.

Interestingly, across all clusters, the data indicates a popularity of non-diabetic individuals, with percentages of 64.7%, 69.5%, 70.7%, and 94%, respectively. This suggests that the majority of individuals in the dataset are in good health when it comes to their diabetic status.

When considering the "physical activity" variable, it is clearly shows that all clusters demonstrate a predominance of individuals engaged in physical activities, as indicated by higher rates of "Physical Activity = Yes" with percentages of 54.6%, 75.8%, 55.5%, and 85.8%, respectively. This also indicates that individuals in the dataset appear to be in good physical shape. However, it's worth noting that this variable remains crucial when seeking to identify factors contributing to the risk of heart disease.

Cluster 4 emerges as unique cluster, with the highest percentage of non-smokers, standing at 71%, and a minority of 29% being smokers. Conversely, the other three clusters display a higher percentage of smokers, with rates of 54%, 68.4%, and 54.7%, as evident in the mean statistics. Furthermore, we investigated variable significance, a critical aspect discussed in the Ward Clustering section. Consistently, BMI emerges as the most important variable with a value of 1, indicating its vital role in determining the risk of heart disease. But in this cluster, "physical activity" registers as the least significant variable, with a value of 0.067664. The "gender" variable appears to hold no relevance in both Ward Clustering and 90% Ward Clustering. These findings contribute to a more comprehensive understanding of the factors affecting heart disease risk.

When utilizing the main dataset of Ward Clustering, three distinct clusters were identified, comprising a total of 131,415 data values. In contrast, 90% sample dataset of Ward Clustering resulted in the formation of four clusters with a total of 118,274 data values. These clusters are considered most stable, as it is generally considered stable when the cluster count is nearly the same or within a difference of approximately +/- 2. Therefore, we can positively say that the clusters are stable in the Ward Clustering. Below attached are the clusters for both the main and 90% sample set.

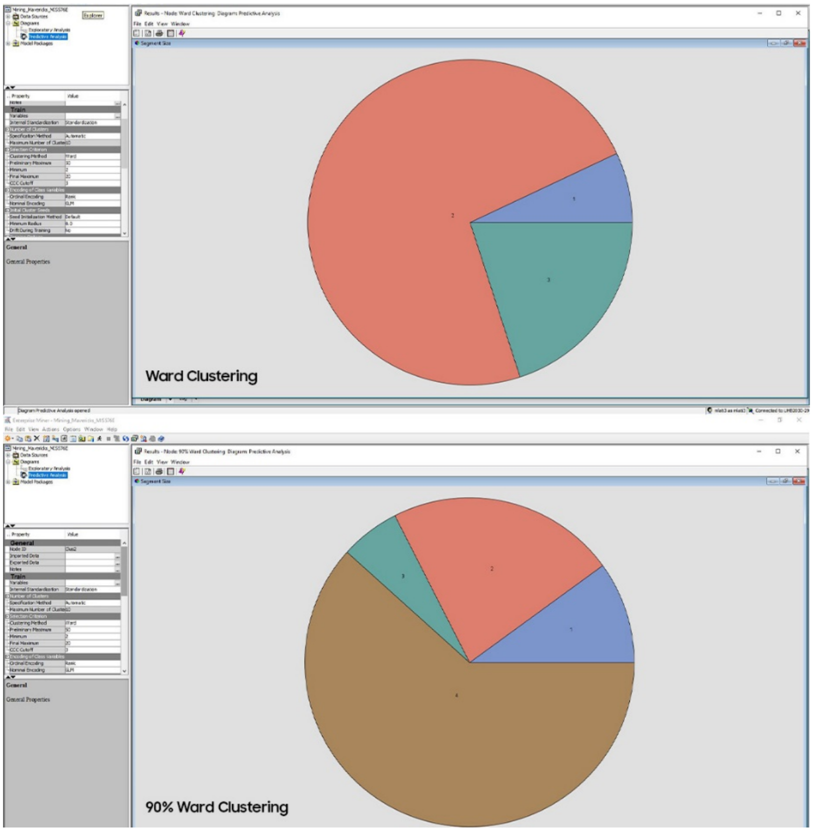

Figure-1: Clusters distribution in ward clustering and 90% ward clustering

In the end, despite minor variations in variable values across clusters between the main dataset and the 90% sample dataset, both datasets exhibit only a one-cluster difference as we know the clusters count is nearly the same or within a difference of approximately +/- 2 are considered to be stable. This reinforces our confidence in the stability of the clusters generated through Ward Clustering. As a result, these cluster insights provide valuable information for assessing the risk of heart disease, with *BMI* emerging as a significant variable in both datasets.

Predictive Analysis (supervised data mining techniques):

Predictive analysis is a data-driven methodology that uses statistical algorithms and historical data to forecast future trends or events. Based on patterns and relationships found in historical data, it seeks to predict outcomes, behaviors, or trends. Predictive analysis is used in various fields to enhance the data driven decisions and optimized the processes (Davenport, T. H., Harris, J., & Shapiro, J., 2010). We did predictive analysis using different supervised techniques like Classification/Decision Trees, Logistic Regression, and Neural Networks. Within these techniques we used different models with variations in their properties to access the best model among the techniques.